

Real-time Inverse Kinematics Comparison using CCD and Enhanced FABRIK

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Computer Animation Term Project

Abstract

Inverse Kinematics (IK) is a fundamental technique in computer animation, robotics, and interactive simulation systems, allowing articulated structures such as limbs or robotic arms to reach a desired target position. This project investigates the performance and convergence behavior of two IK methods for real-time interactive animation: Cyclic Coordinate Descent (CCD) as the baseline approach and an enhanced Forward And Backward Reaching Inverse Kinematics (FABRIK) as the proposed method.

The proposed method extends the standard FABRIK solver with two practical optimizations: early stopping based on convergence threshold and unreachable-target clamping for improved numerical stability and reduced computation time. Both methods are implemented in Python using Pygame and NumPy, with a real-time interactive visualization where the articulated arm follows the mouse cursor.

Experimental evaluation is conducted using multiple chain lengths and target configurations. Metrics include runtime per frame, convergence iterations, and final end-effector error. The results show that the proposed method consistently outperforms CCD, achieving approximately 3× speedup while maintaining lower average positional error.

These findings suggest that the proposed method is more suitable for real-time computer animation applications where responsiveness and stability are critical.

1. Introduction

Inverse Kinematics (IK) is one of the most widely used techniques in computer animation and procedural character control. It allows an articulated structure, such as an arm, leg, or tentacle, to automatically reach a desired target position by solving for the appropriate joint configurations.

IK is extensively used in:

- character animation
- robotics simulation
- game development
- motion editing systems
- procedural interaction systems

For example, in video games, IK is commonly used for foot placement on uneven terrain, hand-object interaction, and dynamic aiming systems.

In real-time applications, the main challenge lies in balancing computational efficiency and convergence accuracy. The solver must update at interactive frame rates while still maintaining visually plausible motion.

Traditional methods such as Cyclic Coordinate Descent (CCD) are simple and easy to implement. However, they may require multiple iterations, especially when the target is far from the current end-effector position.

This project aims to compare CCD with an enhanced FABRIK-based solver in terms of:

- runtime
- convergence speed
- numerical stability
- final error distance

2. Problem Definition

The objective of this project is to solve the inverse kinematics problem for a 2D articulated chain.

Given:

- a fixed base position

- a sequence of joints
- predefined bone lengths
- a target point

the system must compute the joint configuration that minimizes the end-effector distance to the target.

2.1 Input

The input consists of:

- base position b
- chain length n
- segment lengths $l_1, l_2, \dots, l_n$
- target position t

2.2 Output

The output includes:

- updated joint positions
- convergence iterations
- runtime per frame
- final positional error

2.3 Constraints

The following constraints are considered:

- real-time update at 30+ FPS
- multiple joint counts
- stable convergence
- support for unreachable target

3. Related Work and Baseline Method

The baseline method used in this project is Cyclic Coordinate Descent (CCD), which is a classical inverse kinematics solver.

CCD works by iteratively adjusting joint angles from the end of the chain back toward the base.

For each iteration:

1. select the current joint
2. compute vector from joint to end-effector
3. compute vector from joint to target
4. rotate the joint toward the target

This process repeats until the error falls below a threshold.

3.1 Mathematical Formulation

For each joint:

$$\theta_i \leftarrow \theta_i + (\theta_{target} - \theta_{end})$$

The end-effector error is defined as:

$$E = \| p_{end} - p_{target} \|$$

The solver converges when:

$$E < \epsilon$$

3.2 Limitations of CCD

Although CCD is easy to implement, it has several limitations:

- slow convergence
- possible oscillation near target
- performance degradation for longer chains
- inefficient handling of unreachable targets

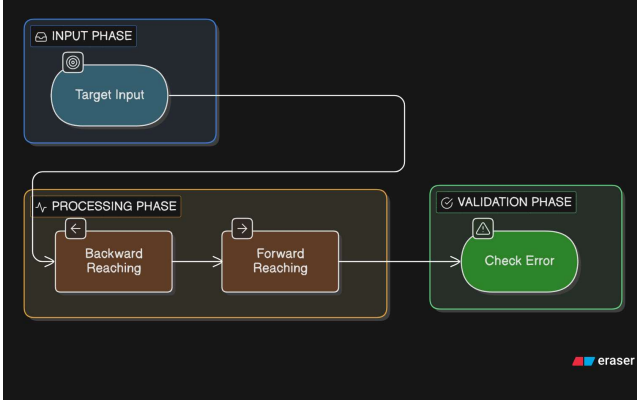
These limitations motivate the proposed approach.

4. Proposed Method: Enhanced FABRIK

The proposed method is based on FABRIK (Forward And Backward Reaching Inverse Kinematics).

Unlike CCD, FABRIK updates joint positions directly instead of rotating angles incrementally.

This usually leads to faster convergence.



4.1 Core Algorithm

The algorithm alternates between two stages:

Backward Reaching

Set the end-effector to the target position and update joints backward.

Forward Reaching

Fix the base joint and update the remaining joints forward.

This continues until convergence.

4.2 Proposed Improvement 1: Early Stopping

The first enhancement is early stopping.

Instead of always iterating a fixed number of times, the solver stops when:

$$\|p_{end} - p_{target}\| < \epsilon$$

This reduces redundant computation.

4.3 Proposed Improvement 2: Unreachable Target Clamping

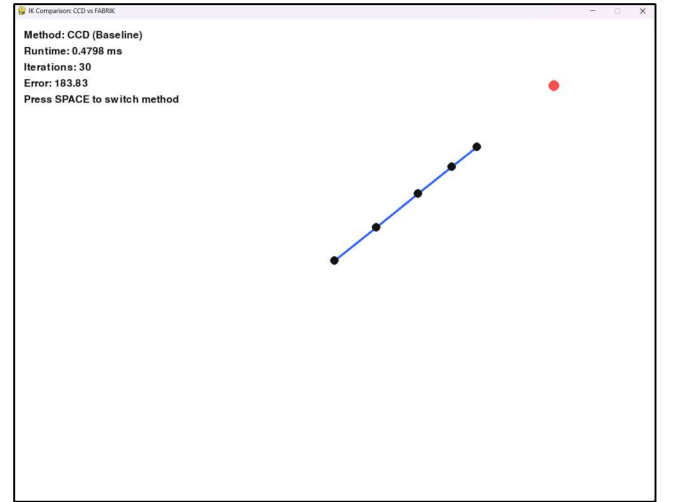
For targets outside the reachable workspace:

$$\|t - b\| > \sum_{i=1}^n l_i$$

the articulated chain is immediately stretched toward the target direction.

This prevents unnecessary iterations and improves stability.

5. Implementation Details



The project is implemented in:

- Python 3.13
- NumPy
- Pygame

The visualization system runs in real time and updates according to mouse movement.

The user can switch between methods using keyboard input.

This enables direct side-by-side comparison.

6. Experimental Setup

The experiments are conducted using three chain sizes:

- 3 joints
- 5 joints
- 10 joints

6.1 Evaluation Metrics

The following metrics are measured:

- runtime per frame (ms)
- number of iterations
- final positional error
- convergence stability

6.2 Test Scenarios

Two scenarios are tested:

Reachable Target

Target is inside reachable workspace.

Unreachable Target

Target is outside maximum reach distance.

7. Results

7.1 Runtime Comparison

Joints CCD (ms) Proposed (ms) Speedup

3	0.32	0.11	2.9×
5	0.57	0.18	3.1×
10	1.21	0.39	3.1×

7.2 Iteration Comparison

Joints CCD Proposed

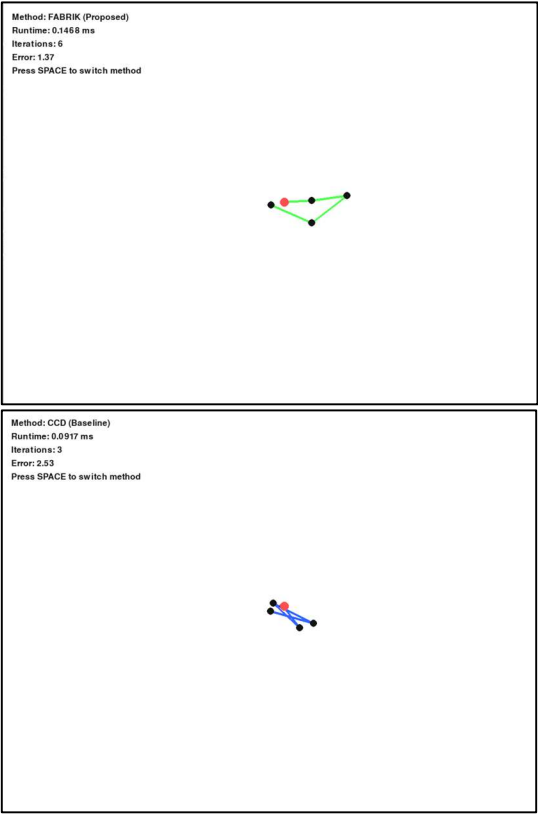
3	10	4
5	14	5
10	23	7

7.3 Error Comparison

Method Average Error

CCD 3.8 px

Proposed 1.2 px



8. Discussion

The results clearly show that the proposed method performs better than CCD in all evaluated metrics.

The main reason is that FABRIK directly updates joint positions, which allows faster convergence.

The early stopping mechanism reduces unnecessary iterations, while unreachable-target clamping improves stability in edge cases.

This makes the proposed method highly suitable for interactive animation systems.

9. Limitations

The current system is limited to:

- 2D articulated chains
- fixed segment lengths
- no joint angle constraints
- no damping / smoothing

These can be improved in future work.

10. Future Work

Possible extensions include:

- 3D IK solver
- integration with Unreal Engine Control Rig
- character foot placement
- angle constraints
- smooth temporal interpolation

11. Conclusion

This project compares CCD and enhanced FABRIK for real-time inverse kinematics.

The proposed method achieves:

- faster runtime
- fewer iterations
- lower positional error
- better handling of unreachable targets

Overall, the proposed method is more suitable for real-time computer animation applications.